

Research Design

Chapter 3

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 - (Pure and Applied Research , Exploratory or Formulative Research
Descriptive Research , Diagnostic Research , Evaluation Studies, Action
Research , Experimental Research, Analytical Study or Statistical
Method, Historical Research, Surveys, Case Study, Field Studies)

Meaning of Research Design

- Research design is the framework of research methods and techniques chosen by a researcher to conduct a study.
- The design allows researchers to sharpen the research methods suitable for the subject matter and set up their studies for success.

- Decisions regarding what, where, when, how much, by what means concerning an inquiry or a research study constitute a research design.
- “A research design is the arrangement of conditions for collection and analysis of data in a manner that aims to combine relevance to the research purpose with economy in procedure.”
- Research design is the conceptual structure within which research is conducted;
 - it constitutes the blueprint for the collection, measurement and analysis of data.
- Research design includes an outline of what the researcher will do from writing the hypothesis and its operational implications to the final analysis of data

- **Research design elements**

- ***Clear purpose:*** The research question or hypothesis must be clearly defined and focused.
- ***Sampling:*** This includes decisions about sample size, sampling method, and criteria for inclusion or exclusion. The approach varies for different research design types.
- ***Data collection:*** This research design element involves the process of gathering data or information from the study participants or sources. It includes decisions about what data to collect, how to collect it, and the tools or instruments that will be used.
- ***Data analysis:*** All research design types require analysis and interpretation of the data collected. This research design element includes decisions about the statistical tests or methods that will be used to analyze the data, as well as any potential confounding variables or biases that may need to be addressed.

- ***Type of research methodology:*** This includes decisions about the overall approach for the study.
- ***Time frame:*** An important research design element is the time frame, which includes decisions about the duration of the study, the timeline for data collection and analysis, and follow-up periods.
- ***Ethical considerations:*** The research design must include decisions about ethical considerations such as informed consent, confidentiality, and participant protection.
- ***Resources:*** A good research design takes into account decisions about the budget, staffing, and other resources needed to carry out the study.

- The overall research design into the following parts:
 - **the sampling design** which deals with the method of selecting items to be observed for the given study
 - **the observational design** which relates to the conditions under which the observations are to be made;
 - **the statistical design** which concerns with the question of how many items are to be observed and how the information and data gathered are to be analysed;
 - **the operational design** which deals with the techniques by which the procedures specified in the sampling, statistical and observational designs can be carried out.

- The important features of a research design as under:
 - It is a plan that specifies the sources and types of information relevant to the research problem.
 - It is a strategy specifying which approach will be used for gathering and analyzing the data.
 - It also includes the time and cost budgets.

- Thus a research design must contain
 - Accurate purpose statement
 - Techniques to be implemented for collecting and analyzing research
 - The method applied for analyzing collected details
 - Type of research methodology
 - Probable objections to research
 - Settings for the research study
 - Timeline
 - Measurement of analysis

Need for research design

- Facilitates the smooth sailing of the various research operations, thereby making research as efficient as possible yielding maximal information with minimal expenditure of effort, time and money.
 - (similar to a blueprint of a house, before its actually constructed)

- Provides for advance planning of the methods to be adopted for collecting the relevant data and the techniques to be used in their analysis, keeping in view the objective of the research and the availability of staff, time and money
- The design helps the researcher to organize his ideas in a form whereby it will be possible for him to look for flaws and inadequacies.

Features of a good design

- A good research design minimizes bias and maximizes the reliability of the data collected and analyzed is considered a good design.
- A good design gives the smallest experimental error.
- A design which yields maximal information and provides an opportunity for considering many different aspects of a problem is considered most appropriate and efficient design in respect of many research problems.

- Characteristics of a good research design:

Characteristics of research design



Neutrality



Reliability



Validity



Generalization

- ***Neutrality***: Right from the study assumptions to setting up the study, a neutral stance must be maintained, free of pre-conceived notions.
(neutral & unbiased results)
 - The researcher's expectations or beliefs should not color the findings or interpretation of the findings.
- ***Reliability***: A reliable research design must allow for results to be consistent, with few errors due to chance.
- ***Validity***: Validity refers to the minimization of nonrandom (systematic) errors.
 - A good research design must employ measurement tools that ensure validity of the results.

- ***Generalizability:*** The outcome of the research design should be applicable to a larger population and not just a small [sample](#).
 - A generalized method means the study can be conducted on any part of a population with similar accuracy.
- ***Flexibility:*** A research design should allow for changes to be made to the research plan as needed, based on the data collected and the outcomes of the study

- For exploratory type of research □ the major emphasis is on discovery of ideas and insights, the research design most appropriate must be flexible enough to permit the consideration of many different aspects of a phenomenon.
- When the research is descriptive type □ relates to an accurate description of a situation or of an association between variables, accuracy becomes a major consideration.
 - a research design which minimises bias and maximises the reliability of the evidence collected is considered a good design.

- Studies involving the testing of a hypothesis of a causal relationship between variables
 - □ requires a design which will permit inferences about causality in addition to the minimization of bias and maximization of reliability

Benefits of research design

- **Clarity of research objectives:** Research design provides a clear understanding of the research objectives and the desired outcomes.
- **Increased validity and reliability:** To ensure the validity and reliability of results, research design help to minimize the risk of bias and helps to control extraneous variables.
- **Improved data collection:** Research design helps to ensure that the proper data is collected and data is collected systematically and consistently.

- **Better data analysis:** Research design helps ensure that the collected data can be analyzed effectively, providing meaningful insights and conclusions.
- **Improved communication:** A well-designed research helps ensure the results are clean and influential within the research team and external stakeholders.
- **Efficient use of resources:** reducing the risk of waste and maximizing the impact of the research, research design helps to ensure that resources are used efficiently

Important concepts relating to research design

- Dependent and Independent variables ☐
 - Continuous, discrete, dependent, independent
 - Qualitative & Quantitative research types
- Extraneous variable ☐ Independent variables that are not related to the purpose of the study, but may affect the dependent variable are termed as extraneous variables.
 - Eg : hypothesis that there is a relationship between a child's knowledge in Social sciences & his socio-economic class. Here knowledge is dependent and Socio-economic class independent, but Intelligence of the child may as well affect the knowledge, even not directly related to this hypothesis (study of the researcher)

- The effect noticed on dependent variable as a result of extraneous variable(s) is technically described as an ‘experimental error’.
 - A study must always be so designed that the effect upon the dependent variable is attributed entirely to the independent variable(s), and not to some extraneous variable or variables.
- Control □ One important characteristic of a good research design is to minimize the influence or effect of extraneous variable(s).
 - The technical term ‘control’ is used when we design the study minimizing the effects of extraneous independent variables.

- Confounded relationship □ When the dependent variable is not free from the influence of extraneous variable(s), the relationship between the dependent and independent variables is said to be confounded by an extraneous variable(s)
- Research hypothesis □ When a prediction or a hypothesized relationship is to be tested by scientific methods, it is termed as research hypothesis.
 - The research hypothesis is a predictive statement that relates an independent variable to a dependent variable. Usually a research hypothesis must contain, at least, one independent and one dependent variable

- Experimental and non-experimental hypothesis-testing research □
 - When the purpose of research is to test a research hypothesis, it is termed as hypothesis-testing research.
 - It can be of the experimental design or of the non-experimental design.
 - Research in which the independent variable is manipulated is termed **‘experimental hypothesis-testing research’**
 - A research in which an independent variable is not manipulated is called **‘non-experimental hypothesis-testing research’**
 - Eg : Experiment relating to checking if intelligence affect the reading ability of students . **Non-experimental**, when a group of students taken, their intelligence and reading ability measured ,and compared using correlation score.
 - **Experimental**, when the group is divided randomly into 2 subgroups. One subgroup given extra training first, and then their reading ability measured; whereas the other subgroup no training given before measuring.

- Experimental and control groups □ In an experimental hypothesis-testing research when a group is exposed to usual conditions, it is termed a '**control group**', but when the group is exposed to some novel or special condition, it is termed an '**experimental group**'.
 - Eg : In medical, testing the effect of a new medicine for a disease; create two groups – one group administered the new medicine (experimental group), the other group not given medicine (control group)
 - It is possible to design studies which include only experimental groups or studies which include both experimental and control groups

- Treatments □ The different conditions under which experimental and control groups are put are usually referred to as ‘treatments’.
 - if we want to determine through an experiment the comparative impact of three varieties of fertilizers on the yield of wheat, in that case the three varieties of fertilizers will be treated as three treatments.

- Experiment □ The process of examining the truth of a statistical hypothesis, relating to some research problem, is known as an experiment.
 - For example, we can conduct an experiment to examine the usefulness of a certain newly developed drug.
- Experiments can be of two types
 - **Absolute experiment** □ If we want to determine the impact of a fertilizer on the yield of a crop, it is a case of absolute experiment
 - **Comparative experiment** □ if we want to determine the impact of one fertilizer as compared to the impact of some other fertilizer, it's a case of comparative experiment

- **Experimental unit(s)** □ The pre-determined plots or the blocks, where different treatments are used, are known as experimental units.

Different research designs

- Different research designs can be conveniently described if we categorize them as:
 - 1. research design in case of exploratory research studies
 - 2. research design in case of descriptive and diagnostic research studies
 - 3. research design in case of hypothesis-testing research studies.

- Exploratory Research
 - Also termed as formulative research studies.
 - The main purpose □ formulating a problem for more precise investigation
 - The major emphasis in such studies is on the discovery of ideas and insights.
 - Here the research design must be flexible enough to provide opportunity for considering different aspects of a problem under study.

- The following three methods in the context of research design are talked about:
 - the survey of concerning literature; □
 - The most simple and fruitful method of formulating precisely the research problem or developing hypothesis.
 - Hypotheses stated by earlier workers may be reviewed and their usefulness be evaluated as a basis for further research.
 - It may also be considered whether the already stated hypotheses suggest new hypothesis.
 - The researcher should review and build upon the work already done by others. but in cases where hypotheses have not yet been formulated. his task is to review the available material for deriving the relevant hypotheses from it

- the experience survey □
 - are surveys of people who have had practical experience with the problem to be studied.
 - The object of such a survey is to obtain research insights into the relationships between variables and new ideas relating to the research problem.
 - For such a survey, people who are competent and can contribute new ideas may be carefully selected as respondents to ensure a representation of different types of experience.
 - The respondents so selected may then be interviewed by the investigator.

— **the analysis of in sight- stimulating examples** □

- It is particularly suitable in areas where there is little experience to serve as a guide.
- This method consists of the intensive study of selected instances of the phenomenon in which one is interested.
- For this purpose the existing records, if any may be examined, the unstructured interviewing may take place, or some other approach may be adopted.
- The attitude of the investigator, the intensity of the study, and the ability of the researcher to draw together diverse information into a unified integration are the main features which make this method an appropriate procedure for evoking insights.
- A few examples of in sight- stimulating cases are the reactions of strangers, the reactions of marginal individuals, the study of individuals who are in transition from one stage to another, the reactions of individuals from different social strata, and the like.

- **Examples of exploratory research in CS**

- Exploratory research in computer science involves investigating areas that are not yet well-understood or are emerging.
 - Quantum Computing Algorithms: Exploring new algorithms that could leverage the principles of quantum mechanics to solve problems more efficiently than classical computers.
 - Artificial General Intelligence (AGI): Researching the development of AI systems with generalized cognitive abilities, aiming to create machines that can perform any intellectual task that a human can.
 - Neuromorphic Computing: Investigating computing systems designed to mimic the neural structures and processes of the human brain to improve efficiency and adaptability.

- Edge Computing Architectures: Examining new architectures and methods for performing computations closer to data sources, which can enhance performance and reduce latency in IoT systems.
- Explainable AI (XAI): Developing techniques to make AI decisions and processes more transparent and understandable to humans, especially in critical areas like healthcare and finance

- Human-Computer Interaction (HCI) in Virtual and Augmented Reality: Exploring new ways to interact with virtual and augmented environments to improve user experiences and applications.
- Blockchain Beyond Cryptocurrencies: Investigating the use of blockchain technology for applications beyond digital currencies, such as supply chain management, voting systems, and secure data sharing.

Research Design in ' Descriptive and Diagnostic Research Studies

- Descriptive research studies are those studies which are concerned with describing the characteristics of a particular individual, or of a group
- Diagnostic research studies determine the frequency with which something occurs, its association with something else, correlation between variables etc.
- Studies concerned with specific predictions. with the narration of facts and characteristics concerning individuals, groups, or situations are all examples of descriptive research studies

- From the point of view of the research design, the descriptive, as well as diagnostic studies share common requirements,
- In descriptive, as well as diagnostic studies, the researcher must be able to define clearly,
 - what is to be measured,
 - find adequate methods for measuring it,
 - A clear cut definition of the population he wants to study.
- The research design must make enough provision for protection against bias, and must maximize reliability, with due concern for the economically completion of the research study.

- The design in such studies must be rigid and not flexible, and must focus attention on the points that follow.
 - a) Formulating the objectives of the study (what the study is about, and why is it being made.)
 - b) Designing the methods of data collection (what techniques of gathering data will be adopted?)
 - c) Selecting the sample (how much material will be needed?)
 - d) Collecting the data (where can the required data be found, and with what time period should the data be related?)
 - e) Processing and analyzing the data.
 - f) Reporting the findings.

- **Descriptive research examples in CS:**
- Descriptive research in computer science focuses on documenting and describing phenomena, systems, or processes as they are, without necessarily exploring causal relationships or underlying mechanisms.
 - **Survey of Programming Languages:** Documenting the features, syntax, and use cases of various programming languages, and how they are adopted across different industries.
 - **Analysis of Software Development Methodologies:** Describing different software development methodologies (e.g., Agile, Scrum, Waterfall) and their adoption patterns, benefits, and limitations in various organizational contexts.
 - **Database Management Systems (DBMS) Comparison:** Cataloging and comparing different DBMSs in terms of their features, performance, scalability, and suitability for different types of applications.
 - **User Experience (UX) Studies:** Describing user interactions and experiences with specific software or hardware systems, including usability testing results and user satisfaction metrics.

- **Network Traffic Analysis:** Documenting patterns and characteristics of network traffic in different environments, such as corporate networks or public Wi-Fi hotspots, to understand usage patterns and performance metrics.
- **Survey of Cybersecurity Threats:** Compiling and describing various types of cybersecurity threats and vulnerabilities, including common attack vectors, threat actors, and their impact on organizations.
- **Review of Cloud Computing Services:** Providing an overview of different cloud service providers, their offerings, pricing models, and how they are used by different sectors.
- **Analysis of Algorithm Performance:** Documenting the performance characteristics of various algorithms under different conditions, including time and space complexity, and comparing their efficiency.

- **Study of Software Bugs and Defects:** Cataloging common types of software bugs and defects, their causes, and their impact on software quality and maintenance.
- **Exploration of Computing Paradigms:** Describing different computing paradigms, such as parallel computing, distributed computing, and grid computing, including their key concepts and typical applications.
- These examples involve gathering and presenting data and insights about various aspects of computer science, providing a comprehensive understanding of current practices, technologies, and trends.

- **Diagnostic research examples in CS**

- Diagnostic research in computer science aims to identify the causes or factors contributing to specific problems or phenomena. It involves investigating issues to understand their origins and impacts.
- diagnostic research aims to uncover the root causes of problems in computer science systems and processes, with the goal of finding effective solutions and improving overall performance and reliability.

- **Debugging Software Bugs:** Analyzing the root causes of software bugs or defects to determine why they occur and how they can be fixed or prevented in the future.
- **Performance Bottlenecks in Systems:** Diagnosing performance issues in software or hardware systems, such as identifying why an application is running slowly or why a server is experiencing high latency.
- **Security Vulnerabilities in Applications:** Investigating the underlying causes of security vulnerabilities in software applications, such as flaws in authentication mechanisms or weaknesses in encryption protocols.
- **Network Failures and Downtime:** Identifying the causes of network failures or outages, including hardware malfunctions, configuration errors, or external attacks, and developing strategies to prevent future occurrences.
- **User Experience Issues:** Diagnosing the causes of poor user experience in software or websites, such as navigation problems, usability issues, or design flaws, and providing solutions for improvement.

- **Algorithmic Errors:** Investigating why a particular algorithm is producing incorrect results or failing to perform as expected, including issues related to algorithm design or implementation errors.
- **Data Integrity Problems:** Identifying the sources of data corruption or inconsistency in databases, including issues with data entry, storage, or transmission processes.
- **Hardware Failures:** Analyzing the causes of hardware failures, such as malfunctioning components or system overheating, to understand the underlying issues and improve hardware reliability.
- **Software Compatibility Issues:** Diagnosing problems related to software compatibility, such as why a particular application does not work correctly on certain operating systems or hardware configurations.
- **Machine Learning Model Performance:** Investigating why a machine learning model is underperforming or producing biased results, including issues with data quality, feature selection, or model training.

Hypothesis Testing or Experimental Research Design

- These are the design where the researcher tests the hypotheses of causal relationships between variables.
- Such studies require procedures that will not only reduce bias and increase reliability, but also permit drawing inferences about causality.
- Usually, experiments meet these requirements. Hence, these are better known as experimental research designs.

- Professor R.A. Fisher enumerated three principles of experimental designs.
 - i) **The Principle of Replication** □
 - Replication refers to the deliberate repetition of an experiment, using nearly identical procedures, which may sometimes be with a different set of subjects in a different setting, and, at distant time periods.
 - It helps to revalidate a previous study, or to raise some questions about the previous studies.
 - As each treatment is applied in many experimental units instead of one. the statistical accuracy of the experiments is increased.

- Examples of Replication control principle
 - Crop productivity
 - If you want to test the productivity of a crop with different types of seeds, you can randomly assign each seed to a unit of land and repeat the process so that each unit gets each seed at least once.
 - School sports grading
 - You can ask students to repeat the same sports-related activities to measure their skills.
 - Diet and weight loss
 - You can repeat a study on the effect of a specific diet on weight loss using the same diet plan and measurement methods.

– The Principle of Randomization:

- A statistical process that involves assigning items or subjects to different groups in a way that each item or subject has an equal chance of being assigned to any group.
- Randomization refers to a technique in which each member of the population, or, universe has an equal and independent chance of being selected.
- This provides for random distribution of the effects of unknown or unspecified extraneous variables. over different groups. thus, balancing their effects to a great extent.
- This is a method of controlling the extraneous variables and reducing experimental error.

- Examples

- Simple randomization □ a basic method of randomization that involves flipping a coin to determine the assignment of each participant.
 - For example, if there are two treatment groups, heads could represent the control group and tails could represent the treatment group.
- Block randomization □ A method that involves randomizing participants into groups that have equal sample sizes. This method helps to ensure a balance in sample size across groups over time.
- Randomized block design □ An experimental design that involves grouping subjects into blocks and then randomly assigning different treatments to the units in each block. This method can help researchers account for potentially unwanted variables

– Experimental matched design

- A method that involves blocking and then matching up individual participants from each block based on specific characteristics.
- Within each matched pair or group, participants are randomly assigned to one of the conditions in the experiment and their outcomes are compared.

- **The principle of local control**
 - a technique that aims to increase experimental precision by controlling external variables.
 - It's a more refined technique than replication or randomization, and it's often used in experiments with more than one formation of experimental units.
 - Examples of local control in research:
 - Website design
 - To study how easy it is to navigate an electronics brand's website, researchers can group customers by age to create homogeneous lots.

- Wheat crop experiment
 - If an experiment is conducted in a location with different soil characteristics, researchers can use local control to create four blocks of 12 homogeneous plots.
 - Then, they can randomly assign each plot to one of 12 combinations of fungicide levels and cultivars.
- Local control can also involve grouping experimental units into blocks so that they are homogeneous within the blocks.
- This eliminates variation among the blocks, and the error component should only contain variation due to treatments.
- This can increase efficiency and reduce experimental error.